GNSS Positioning During Reference Data Outages

Observation models, clock propagation and implementation

Technical implementation note

18 October 2026

Abstract

When reference corrections stop arriving but satellite tracking continues, the rover can bridge the outage using old reference observations or time-differenced rover carrier phases. Both approaches retain time-dependent satellite clock, orbit and atmospheric errors. This note derives consistent clock corrections, complete reference-observation prediction and anchored delta-phase positioning. It specifies signs, units, transmission times, covariance handling, pseudocode and numerical verification cases. The goal is an implementable measurement model, not a guarantee that an RTK fixed status remains centimetre-accurate throughout an outage.

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1 Scope and recommended architecture

Assume a stationary reference antenna, continuing GNSS reception at the rover, a reliable last synchronous RTK solution at reception epoch t_0, and a current reception epoch $t_1 > t_0$. The outage age is $a = t_1 - t_0$. Signal loss, changing reference coordinates and changing virtual reference stations require additional handling.

Preserve the last valid solution, observations, ambiguity states, atmosphere states and their joint covariance. Bridge the outage with either an asynchronous observation model or anchored delta-phase processing. Recompute satellite geometry and full satellite clocks at their actual transmission times. Increase model uncertainty as corrections age. Resume ordinary RTK only after validating returned data and ambiguity continuity.

The equations below are derived from the conventional GNSS observation model [1]. Shu et al. investigate asynchronous RTK error contributions [4]; their results support treating time-varying residual clock and atmosphere errors explicitly. Kim et al. address correlations in TDCP-based GPS/INS filtering [5]. Neither paper establishes a universal safe outage duration for a particular receiver.

2 Notation and sign contract

Symbol	Meaning and units
R, B	Rover and reference receiver
s, j	Satellite and reference satellite
t_i	Reception epoch in GNSS system time, seconds
$u^s_{X,i}$	Transmission epoch for receiver X, satellite s, epoch i
P^s_X, L^s_X	Code and carrier observables, both in metres
$L^s_X = \lambda_s \varphi^s_X$	Carrier phase converted from cycles to metres
b_X, b_s	Receiver and satellite clock offsets, seconds
ρ^s_X	Geometric range including Earth rotation
T^s_X, I^s_X	Tropospheric and signal-specific ionospheric delays, metres
B^s_X	Other deterministic carrier corrections, metres
N^s_X	Constant phase ambiguity on a continuous tracking arc
$\widehat{(\cdot)}$	Model or estimate; no hat denotes the true quantity
c	$299\,792\,458\ \mathrm{m\,s^{-1}}$

Define a clock offset as clock reading minus system time:

$$b_s(u) = t_{\mathrm{SV},s}(u) - u, \qquad b_X(t) = t_{X,\mathrm{clock}}(t) - t. \tag{2.1}$$

A positive satellite offset means that the satellite clock is ahead. With one signal and frequency understood,

$$P^s_X(t) = \rho^s_X(t) + c[b_X(t) - b_s(u^s_X)] + T^s_X(t) + I^s_X(t) + B^s_{P,X}(t) + \epsilon^s_{P,X}, \tag{2.2}$$

$$L^s_X(t) = \rho^s_X(t) + c[b_X(t) - b_s(u^s_X)] + T^s_X(t) - I^s_X(t) + B^s_X(t) + \lambda_s N^s_X + \epsilon^s_X. \tag{2.3}$$

These signs follow [1]. The ambiguity sign is fixed by this definition; adapt the input parser if its phase convention differs. Define the non-clock carrier term

$$U^s_X = T^s_X - I^s_X + B^s_X. \tag{2.4}$$

B^s_X can include phase wind-up and antenna corrections in the chosen convention. Avoid counting these terms twice.

The satellite clock contributes $-cb_s$ to the observable. Removing its modelled contribution requires adding $+c\hat{b}_s$.

3 Transmission time and geometric range

For each observation,

$$u^s_{X,i} = t_i - \tau^s_{X,i}. \tag{3.1}$$

For any old/new observation pair,

$$h_s = u^s_1 - u^s_0 = (t_1 - t_0) - (\tau^s_1 - \tau^s_0). \tag{3.2}$$

Thus h_s is satellite-dependent and not exactly the correction age. An initial pseudorange-based transmission estimate can be refined by an existing validated GNSS observation routine. Do not use a raw receiver clock timestamp as system time without handling b_X.

For a geometric light-time iteration, given system reception time t and receiver position $\boldsymbol{x}$, use

$$u^{(k)} = t - \tau^{(k)}, \tag{3.3}$$

$$\boldsymbol{r}^{(k)}_{\mathrm{rot}} = R_z(-\omega_E \tau^{(k)})\, \boldsymbol{r}_s(u^{(k)}), \tag{3.4}$$

$$\rho^{(k)} = \|\boldsymbol{r}^{(k)}_{\mathrm{rot}} - \boldsymbol{x}\|, \qquad \tau^{(k+1)} = \rho^{(k)}/c, \tag{3.5}$$

where the active rotation is

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{3.6}$$

This expresses transmit-epoch ECEF satellite coordinates in receive-epoch ECEF axes. Start with approximately 0.07 s for a GPS MEO satellite or a code-derived estimate; stop when the range change is below the chosen numerical tolerance. More complete transmission-time routines include propagation delays. Apply Earth rotation once, not both here and again as a separate range correction.

Use absolute week-aware times internally. Convert to constellation-specific time arguments at the navigation interface. A modulo-week operation must not hide an expired ephemeris.

4 Satellite clock propagation

4.1 Broadcast model

For the GPS broadcast clock model [2],

$$\hat{b}_s(u) = a_{f0,s} + a_{f1,s} q + a_{f2,s} q^2 + \Delta_{\mathrm{rel},s}(u), \qquad q = u - t_{\mathrm{oc},s}. \tag{4.1}$$

The coefficients have units s, s/s and s/s^2. The second derivative of the polynomial is $2a_{f2}$.

The periodic relativistic term is [3]

$$\Delta_{\mathrm{rel},s}(u) = -\frac{2}{c^2} \boldsymbol{r}_s(u) \cdot \boldsymbol{v}_s(u). \tag{4.2}$$

These are geocentric satellite position and velocity, not receiver-to-satellite vectors. Equivalently, for the GPS broadcast Keplerian model,

$$\Delta_{\mathrm{rel},s} = F e_s \sqrt{A_s} \sin E_s, \qquad F = -\frac{2\sqrt{\mu}}{c^2}. \tag{4.3}$$

Use the constellation's specified constants. With consistent rotating coordinates the scalar product is unchanged because $\boldsymbol{r} \cdot (\boldsymbol{\omega}_E \times \boldsymbol{r}) = 0$. Use one relativistic formulation, not both. A production broadcast implementation should follow its constellation ICD's orbital and clock algorithm.

4.2 Exact difference within a common parameter set

Define the direction *new minus old*:

$$D_s = \hat{b}_s(u_1) - \hat{b}_s(u_0), \qquad h = u_1 - u_0, \quad q_0 = u_0 - t_{\mathrm{oc}}. \tag{4.4}$$

For the polynomial,

$$D_{\mathrm{poly},s} = a_{f1,s}h + a_{f2,s}[(q_0 + h)^2 - q_0^2] \tag{4.5}$$

$$= (a_{f1,s} + 2a_{f2,s}q_0)h + a_{f2,s}h^2. \tag{4.6}$$

The constant coefficient cancels only for the same parameter set. Writing $Q_s(u) = \boldsymbol{r}_s(u) \cdot \boldsymbol{v}_s(u)$ gives

$$D_{\mathrm{rel},s} = -\frac{2}{c^2}[Q_s(u_1) - Q_s(u_0)]. \tag{4.7}$$

Combining both parts,

$$\boxed{D_s = (a_{f1,s} + 2a_{f2,s}q_0)h + a_{f2,s}h^2 + \frac{2}{c^2}[Q_s(u_0) - Q_s(u_1)].} \tag{4.8}$$

Then $\hat{b}_s(u_1) = \hat{b}_s(u_0) + D_s$. A formula with $a_{f1}(u_0 - u_1)$ but the last bracket of Eq. (4.8) mixes opposite differencing directions under Eq. (2.1). Reversing the overall definition reverses *all* terms.

4.3 Why a nearly linear drift is expected

Taylor expansion gives

$$D_s = \dot{\hat{b}}_s(u_0)h + \tfrac{1}{2}\ddot{\hat{b}}_s(u_0)h^2 + O(h^3), \tag{4.9}$$

where, using consistent coordinate derivatives,

$$\dot{\hat{b}}_s = a_{f1,s} + 2a_{f2,s}(u - t_{\mathrm{oc}}) - \frac{2}{c^2}\left(\boldsymbol{v}_s \cdot \boldsymbol{v}_s + \boldsymbol{r}_s \cdot \boldsymbol{a}_s\right). \tag{4.10}$$

Both the polynomial and periodic relativistic term can appear nearly linear over a short interval. A linear position drift alone does not identify the missing term. Evaluate endpoint clocks in software rather than differentiating noisy sampled velocities.

4.4 Constellation and product conventions

For the usual GLONASS broadcast convention,

$$\hat{b}_s(u) = -\tau_{n,s} + \gamma_{n,s}(u - t_{b,s}), \qquad D_s = \gamma_{n,s}h \tag{4.11}$$

within a common valid parameter set and time convention. Its periodic relativistic contribution is already included; do not add Eq. (4.2) again [2, 3]. RINEX fields and native message parameters can use different signs.

Galileo and BeiDou require their own message, time-system, signal-bias and validity handling. Precise clocks and SSR corrections also require a product-specific adapter. Never assume a provider's correction sign, reference clock, issue-of-data or inclusion of relativity. Code group delays are not automatically carrier-phase corrections.

5 Method A: asynchronous reference–rover differences

Let

$$A^s = L_R^s(t_1) - L_B^s(t_0). \tag{5.1}$$

Substituting Eq. (2.3),

$$\begin{aligned} A^s = {} & \rho_R^s(t_1) - \rho_B^s(t_0) + c[b_R(t_1) - b_B(t_0)] \\ & - c[b_s(u_{R,1}^s) - b_s(u_{B,0}^s)] + U_R^s(t_1) - U_B^s(t_0) \\ & + \lambda_s(N_R^s - N_B^s) + \epsilon_R^s(t_1) - \epsilon_B^s(t_0). \end{aligned} \tag{5.2}$$

Define $D_{RB,s} = \hat{b}_s(u_{R,1}^s) - \hat{b}_s(u_{B,0}^s)$. Correct with

$$\boxed{A_{\text{corr}}^s = A^s + cD_{RB,s}.} \tag{5.3}$$

The geometric model remains $\rho_R^s(t_1) - \rho_B^s(t_0)$. Do not substitute a current reference range while retaining an old reference observation.

5.1 Equivalent adjustment of the reference observable

If an interface applies the clock adjustment to the reference side,

$$\boxed{L_{B,\text{adj}}^s(t_0) = L_B^s(t_0) - cD_{RB,s}.} \tag{5.4}$$

Indeed $L_R^s(t_1) - L_{B,\text{adj}}^s(t_0) = A_{\text{corr}}^s$. This is an algebraic, rover-pair-specific adjustment, not a complete synthetic current reference epoch. If inputs are already corrected by $L_X^s + c\hat{b}_s(u_X^s)$, Eq. (5.3) is already present.

5.2 Between-satellite differences

For satellites s, j in a compatible receiver-clock group,

$$A_{\text{corr}}^{s,j} = (A^s - A^j) + c(D_{RB,s} - D_{RB,j}). \tag{5.5}$$

The common receiver clock difference cancels. Fixed double-difference ambiguities can be retained only while arcs and bias conventions remain valid. Across constellations estimate appropriate inter-system offsets. GLONASS FDMA hardware biases and unequal wavelengths need explicit handling; convert observations to metres first.

6 Method B: synthesize a current reference observation

For a stationary reference, difference its model between t_0, t_1:

$$L_B^s(t_1) - L_B^s(t_0) = \Delta\rho_B^s + c\Delta b_B - c\Delta b_s + \Delta U_B^s + \Delta\epsilon_B^s. \tag{6.1}$$

Let $D_{B,s} = \hat{b}_s(u_{B,1}^s) - \hat{b}_s(u_{B,0}^s)$. The predictor is

$$\boxed{\widetilde{L}_B^s(t_1) = L_B^s(t_0) + \Delta\hat{\rho}_B^s + c\Delta\hat{b}_B - cD_{B,s} + \Delta\hat{U}_B^s,} \tag{6.2}$$

where $\Delta\hat{U}_B^s = \Delta\hat{T}_B^s - \Delta\hat{I}_B^s + \Delta\hat{B}_B^s$. No base motion is needed for $\Delta\rho_B^s$ to be large: the satellite moves. The ambiguity cancels only under the assumed continuous base arc. The actual base phase and clock are unavailable, so this is a prediction with correlated errors, not a new measurement.

The base clock increment is common to compatible satellites. In a double-difference engine one may choose a synthetic common-clock gauge, for example $\Delta\hat{b}_B = 0$, and let the clock parameter or differencing absorb it. This is not a physical base-clock measurement.

Rover and synthetic base transmission times at t_1 still differ. Remove the remaining synchronous satellite-clock contribution with

$$[L_R^s(t_1) - \widetilde{L}_B^s(t_1)] + c[\hat{b}_s(u_{R,1}^s) - \hat{b}_s(u_{B,1}^s)]. \tag{6.3}$$

Together with Eq. (6.2), the total clock adjustment is exactly $+cD_{RB,s}$. Exact synchronous cancellation is a short-baseline approximation.

A code predictor uses $+\Delta I_B^s$ instead of $-\Delta I_B^s$ and the appropriate code biases. Do not apply the carrier predictor unchanged to code.

7 Method C: anchored rover delta-phase

7.1 Observation equation

Use two rover observations on the same continuous arc:

$$Z^s = L_R^s(t_1) - L_R^s(t_0). \tag{7.1}$$

Then

$$Z^s = \Delta\rho_R^s + c\Delta b_R - c\Delta b_s + \Delta U_R^s + \Delta\epsilon_R^s, \tag{7.2}$$

since $\lambda_s \Delta N_R^s = 0$. With $D_{R,s} = \hat{b}_s(u_{R,1}^s) - \hat{b}_s(u_{R,0}^s)$ define

$$\boxed{z^s = Z^s + cD_{R,s} - \Delta\hat{U}_R^s.} \tag{7.3}$$

Fit

$$z^s = \rho_R^s(t_1, \boldsymbol{x}_1) - \rho_R^s(t_0, \boldsymbol{x}_0) + \Delta\beta_R + e^s, \qquad \Delta\beta_R = c\Delta b_R. \tag{7.4}$$

The anchor $\boldsymbol{x}_0$ is the last reliable RTK position. Satellite motion remains in the two independently evaluated ranges.

7.2 Position update

Given iterates $\bar{\boldsymbol{x}}_1, \overline{\Delta\beta}$, compute

$$v_s = z^s - \left[\rho_R^s(t_1, \bar{\boldsymbol{x}}_1) - \rho_R^s(t_0, \hat{\boldsymbol{x}}_0) + \overline{\Delta\beta}\right]. \tag{7.5}$$

Let $\boldsymbol{e}_{s,1}$ point from rover to satellite at the current epoch. The usual first-order row is

$$H_s = \begin{bmatrix} -\boldsymbol{e}_{s,1}^T & 1 \end{bmatrix}, \quad \boldsymbol{d} = \begin{bmatrix} \delta\boldsymbol{x}_1 \\ \delta\Delta\beta \end{bmatrix}. \tag{7.6}$$

Solve

$$\min_{\boldsymbol{d}} (\boldsymbol{v} - H\boldsymbol{d})^T C^{-1} (\boldsymbol{v} - H\boldsymbol{d}), \qquad \bar{\boldsymbol{x}}_1 \leftarrow \bar{\boldsymbol{x}}_1 + \delta\boldsymbol{x}_1. \tag{7.7}$$

Use a whitened QR or square-root solver, not an explicit matrix inverse. Recompute transmission times, clocks and geometry until convergence. Four independent observations are the minimum for three coordinates plus one clock increment; redundancy and good geometry are necessary in practice. For exact Jacobians include light-time dependence or differentiate the full observation function.

If the anchor is estimated jointly, the row is approximately

$$H_s = \begin{bmatrix} -\boldsymbol{e}_{s,1}^T & +\boldsymbol{e}_{s,0}^T & 1 \end{bmatrix} \tag{7.8}$$

for $(\delta\boldsymbol{x}_1, \delta\boldsymbol{x}_0, \delta\Delta\beta)$. Give $\boldsymbol{x}_0$ its RTK prior and preserve cross-covariances with stored measurements and states.

7.3 Relationship to old-reference processing

The exact raw-observation identity is

$$\boxed{L_R^s(t_1) - L_B^s(t_0) = [L_R^s(t_1) - L_R^s(t_0)] + [L_R^s(t_0) - L_B^s(t_0)].} \tag{7.9}$$

The formulations rearrange the same data when anchor information is preserved. Replacing the full anchor state by an error-free coordinate loses information and changes the statistics. Different filters, satellite selection or additional sensors can produce different results.

Successive short increments do not eliminate a systematic clock error:

$$\sum_{k=1}^{n} [b_s(u_k) - b_s(u_{k-1})] = b_s(u_n) - b_s(u_0). \tag{7.10}$$

Delta-phase can connect smoothly to its anchor while absolute error grows.

8 Residual errors and uncertainty

8.1 Atmosphere propagation required by the observation predictor

The clock equations do not determine the atmospheric increments in Eqs. (6.2) and (7.3). One practical state model uses zenith hydrostatic and wet delays with elevation-dependent mapping functions:

$$\hat{T}_X^s(t) = m_h(e_X^s(t))\hat{Z}_{h,X}(t) + m_w(e_X^s(t))\hat{Z}_{w,X}(t). \tag{8.1}$$

Even a constant zenith delay produces a changing slant delay as elevation changes. Predict the zenith states, evaluate the mapping functions at both epochs and subtract the resulting slant delays. Mapping-function arguments such as latitude, height and season are suppressed here, not assumed absent.

For a signal of frequency f, a first-order ionosphere state at frequency $f_\star$ scales as

$$\hat{I}_f^s(t) = \left(\frac{f_\star}{f}\right)^2 \hat{I}_{f_\star}^s(t). \tag{8.2}$$

For a short outage, a slant-state predictor may be $\hat{I}_{f_\star}^s(t_1) = \hat{I}_{f_\star}^s(t_0) + \hat{\dot{I}}_{f_\star}^s a$. Use zero rate only as an explicit persistence assumption and increase its uncertainty. This is a local empirical prediction, not a universal ionosphere law. Changes in receiver position and ray geometry limit its useful duration. Multi-frequency data can support changes in ionospheric states, but introduce their own noise and bias handling. An ionosphere-free carrier combination is

$$L_{\mathrm{IF}} = \frac{f_1^2 L_1 - f_2^2 L_2}{f_1^2 - f_2^2}. \tag{8.3}$$

It removes the first-order ionosphere under the shared-ray model, amplifies noise, and generally loses a simple single integer ambiguity. Its constant ambiguity still cancels in TDCP if both frequency arcs remain continuous. Propagate its covariance using the same linear combination coefficients.

8.2 Clock error after correction

Write $b_s = \hat{b}_s + \eta_s$. After the modelled correction,

$$e_{\mathrm{clk},s} = -c[\eta_s(u_1) - \eta_s(u_0)]. \tag{8.4}$$

If an independently supported residual rate $\hat{\dot{\eta}}_s$ exists, a short-term predictor could use

$$D_{\mathrm{enhanced},s} = D_s + \hat{\dot{\eta}}_s h. \tag{8.5}$$

Single-station residuals also contain orbit, atmosphere and multipath contributions. They do not uniquely identify η_s. This predictor requires validation and an observable state.

An illustrative residual-clock model is

$$\boldsymbol{q} = \begin{bmatrix} \eta \\ \dot{\eta} \end{bmatrix}, \quad F(h) = \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix}, \quad P_1 = FP_0F^T + q_d \begin{bmatrix} h^3/3 & h^2/2 \\ h^2/2 & h \end{bmatrix}. \tag{8.6}$$

Here q_d is the continuous driving-noise intensity for the rate; calibrate it from data. The uncertainty of the increment is

$$\mathrm{Var}(\eta_1 - \eta_0) = h^2(P_0)_{22} + q_d h^3/3. \tag{8.7}$$

Multiply by c^2 for range variance. This model alone does not reveal a particular satellite's error.

8.3 Shared epochs create correlations

If raw scalar phase errors at distinct epochs are independent with variances σ_k^2, fixed-anchor differences $d_k = \epsilon_k - \epsilon_0$ satisfy

$$\mathrm{Var}(d_k) = \sigma_k^2 + \sigma_0^2, \tag{8.8}$$

$$\mathrm{Cov}(d_k, d_l) = \sigma_0^2, \quad k \neq l. \tag{8.9}$$

For adjacent differences $g_k = \epsilon_k - \epsilon_{k-1}$,

$$\mathrm{Cov}(g_k, g_{k+1}) = -\sigma_k^2. \tag{8.10}$$

Repeated use of old base data has the same shared-measurement issue. Use full covariance, a factor graph or augmented/delayed-state filtering rather than repeatedly treating reused data as independent [5].

For a satellite differencing matrix M, propagate

$$\boldsymbol{z}_{\mathrm{DD}} = M\boldsymbol{z}, \qquad C_{\mathrm{DD}} = MCM^T. \tag{8.11}$$

A common reference satellite introduces off-diagonal terms. Model and anchor uncertainty must be included with actual cross-covariances.

9 Implementation interfaces and pseudocode

9.1 Required interfaces

- `geometry(time, position, satellite, nav)` returns range, transmission time, line of sight and validity.
- `clock(tx_time, satellite, nav)` returns full offset in seconds, component terms and product identity.
- `carrier_terms(...)` returns $U = T - I + B$ in metres and uncertainty information.

- The navigation adapter validates health, issue-of-data, time scale, fit interval and bias conventions.
- Record phase units, lock status, slips, half-cycle flags, signal identity and corrections already applied.

Geometry and atmosphere routines are established GNSS primitives; the pseudocode specifies the outage layer, not a complete navigation-message decoder.

9.2 Clock increment

```
function clock_increment(u0, u1, satellite, common_nav):
    require valid_for_both_epochs(common_nav, u0, u1)
    h = absolute_time_difference(u1, u0) # seconds
    q0 = clock_reference_difference(u0, common_nav.toc)
    dpoly = (af1 + 2*af2*q0)*h + af2*h*h
    rel0 = periodic_relativity(u0, satellite, common_nav)
    rel1 = periodic_relativity(u1, satellite, common_nav)
    return dpoly + rel1 - rel0 # seconds
```

This is the GPS-style adapter. GLONASS and precise-product adapters implement their own complete difference. Endpoint evaluation is also valid.

9.3 Anchored delta-phase

```
on_last_valid_synchronous_fix:
    save t0, x0, joint_state_and_covariance
    save rover phases L0[s], arc_id[s], signal_id[s]
    save navigation and correction conventions

on_current_epoch(t1):
    usable = matching_continuous_arcs(current, saved)
    iterate over trial x1 and clock_increment_m:
        for s in usable:
            g0 = geometry(t0, x0, s, consistent_nav)
            g1 = geometry(t1, x1, s, consistent_nav)
            D = full_clock_difference(g0.tx, g1.tx, s)
            dU = carrier_terms(t1, x1, s) \
               - carrier_terms(t0, x0, s)
            z[s] = L1[s] - L0[s] + c*D - dU
            predicted[s] = g1.range - g0.range \
                        + clock_increment_m
            residual[s] = z[s] - predicted[s]
            H[s] = [-g1.line_of_sight, 1]
        construct_joint_covariance_and_priors()
        update = solve_whitened_QR(H, residual, covariance)
        apply_update_and_check_convergence()
    validate_residuals_geometry_arcs_and_uncertainty()
    output position, uncertainty, outage_age, bridge_status
```

For multiple constellations add receiver clock/inter-system states. New satellites lack an old phase on the same arc and cannot simply enter this TDCP equation. Use a separate code/float model or initialize an eligible new anchor with uncertainty.

9.4 Complete reference prediction

```
for each eligible reference satellite s:
    old = geometry(t0, xBase, s, nav)
    new = geometry(t1, xBase, s, nav)
    D = full_clock_difference(old.tx, new.tx, s)
    dU = predicted_reference_terms(t1, s) \
       - reference_terms(t0, s)
    LBasePred[s] = LBaseOld[s] \
                + new.range - old.range \
                + predicted_base_clock_increment_m \
                - c*D + dU
    attach_prediction_covariance_and_shared_anchor_links()
```

An RTK engine must recognize synthetic observations and their correlations. Treating them as independent fresh RTCM observations gives misleading confidence.

10 Validation and operational safeguards

10.1 Deterministic numerical checks

1. **Zero interval:** $u_1 = u_0$ must give $D_s = 0$.
2. **Constant offset:** changing only a_{f0} in a common model must not change D_s.
3. **Linear clock:** $a_{f1} = 5 \times 10^{-12}$, $h = 120$ s, $a_{f2} = 0$ gives $D_{\text{poly}} = 0.6$ ns.
4. **Relativistic increment:** add $D_{\text{rel}} = 0.4$ ns to obtain $D = 1$ ns and $cD = 0.299792458$ m.
5. **Sign:** a raw phase increment containing -0.299792458 m of satellite-clock contribution must be corrected by adding cD.
6. **Quadratic term:** $a_{f1} = 0$, $a_{f2} = 10^{-18}$ s/s^2, $q_0 = 1000$ s, $h = 120$ s gives $D_{\text{poly}} = 2.544 \times 10^{-13}$ s.
7. **Reference adjustment:** verify $L_R - (L_B - cD) = (L_R - L_B) + cD$.
8. **Common mode:** identical D_s must vanish in satellite differences.
9. **Stationary simulation:** with satellite motion and clock rates but noiseless fixed receivers, the full corrected estimator must recover zero displacement.
10. **Boundaries:** test week rollover, navigation updates, receiver clock resets, slips, lost satellites and reference-satellite changes.

10.2 Navigation updates

For different navigation sets,

$$D_s = \hat{b}_s(u_1; \mathcal{E}_1) - \hat{b}_s(u_0; \mathcal{E}_0). \tag{10.1}$$

Constant coefficients no longer cancel. The difference may contain a model discontinuity rather than physical drift. Use a common valid set over a short interval where appropriate; otherwise re-express stored model-dependent states consistently. Handle orbit and clock updates together. Log old/new model differences at a common epoch. Do not impose continuity by an arbitrary offset without accounting for its effect on observations and states.

10.3 Accuracy monitoring and recovery

Monitor horizontal and vertical uncertainty separately, continuity, geometry, innovations and data age. Fixed ambiguities do not certify accurate position during an outage. On recovery validate station identity, coordinates, timestamps, ambiguities and residuals before reusing reference data.

Independent precise correction delivery can extend bridging. NovAtel describes RTK ASSIST using TerraStar corrections and PPP [6]. This is not evidence for the same performance without

incoming corrections. IMU/odometry can constrain motion; confirmed static operation can justify a zero-motion constraint.

11 Interpretation of the supplied experiments

The figures are reproduced from the supplied material. Their original conditions do not establish a receiver-independent guarantee. Figure 1 illustrates successive model improvements. Figure 2 shows similar long-term behaviour for delta-phase and old-reference processing.

The discussed approximate displacement of 0.36 m in 120 s has a mean position-change magnitude

$$0.36/120 = 0.003 \text{ m/s} = 18 \text{ cm/min}. \tag{11.1}$$

As a *range-equivalent scale* only, this is 1.20 ns over the interval or about 10^{-11} s/s. It is not a measured satellite clock rate. For a simple full-rank weighted least-squares model,

$$\delta \boldsymbol{z} = (H^T W H)^{-1} H^T W \boldsymbol{e}, \qquad \boldsymbol{e}_{\text{clk}} = -c\,\Delta\boldsymbol{\eta}. \tag{11.2}$$

Geometry determines the mapping to position and receiver clock; filter priors change it. A nearly linear position curve is compatible with clock-model errors but is not diagnostic by itself.

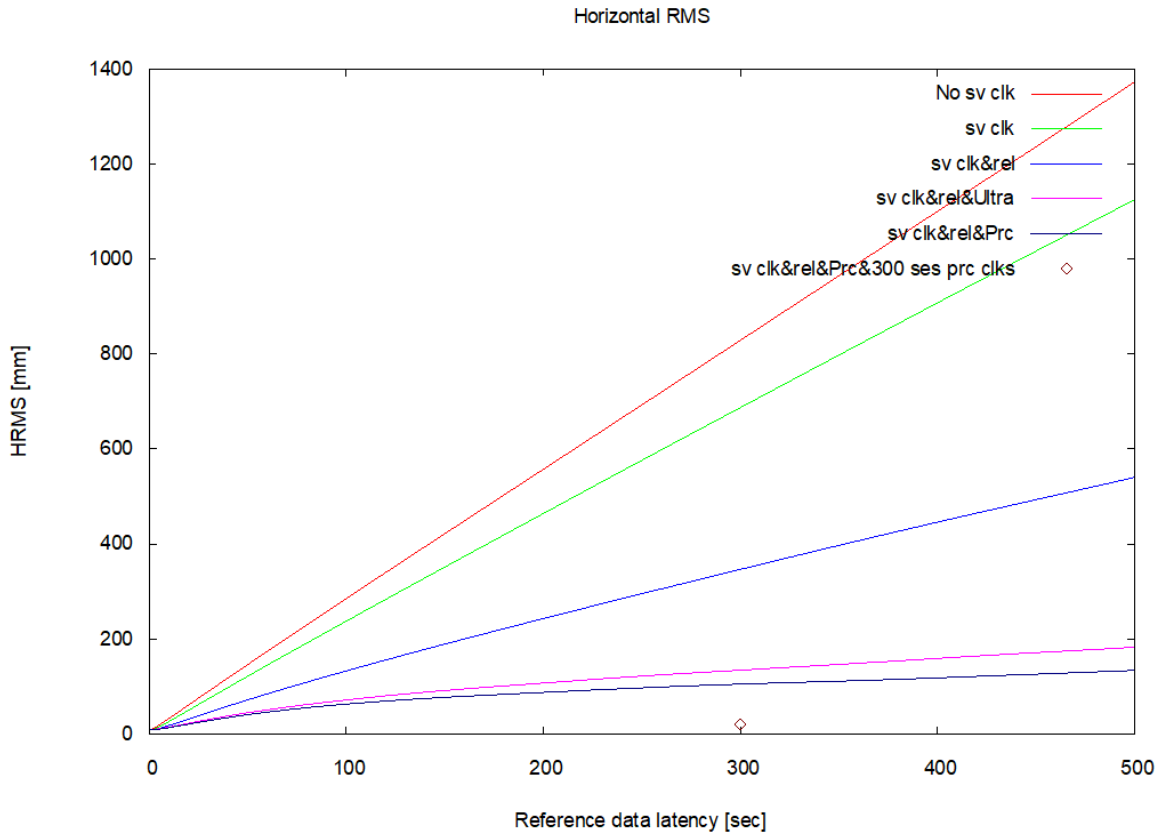


Figure 1: Horizontal RMS degradation for different reference outage time spans and corrections

Figure 1: Historical horizontal RMS versus latency, reproduced from the supplied clock summary [7]. The legend compares omitted clock corrections, clock corrections, additional relativity and improved orbit/clock products. Correcting clocks alone need not remove all drift. This is a historical experiment, not a P23M-specific prediction.

4.1 Delta-Phase Processing

One proposal for keeping the accuracy as long as possible was the so-called delta-phase processing. Here, the epoch-to-epoch differences between the rover data of the last RTK fix and the current rover data epoch are used to compute centimeter-level position differences between the current rover position and the last RTK fix position. Although this looks appealing at first, this method differs from the analyzed “keep the last reference epoch available” only in the order of the operations performed.

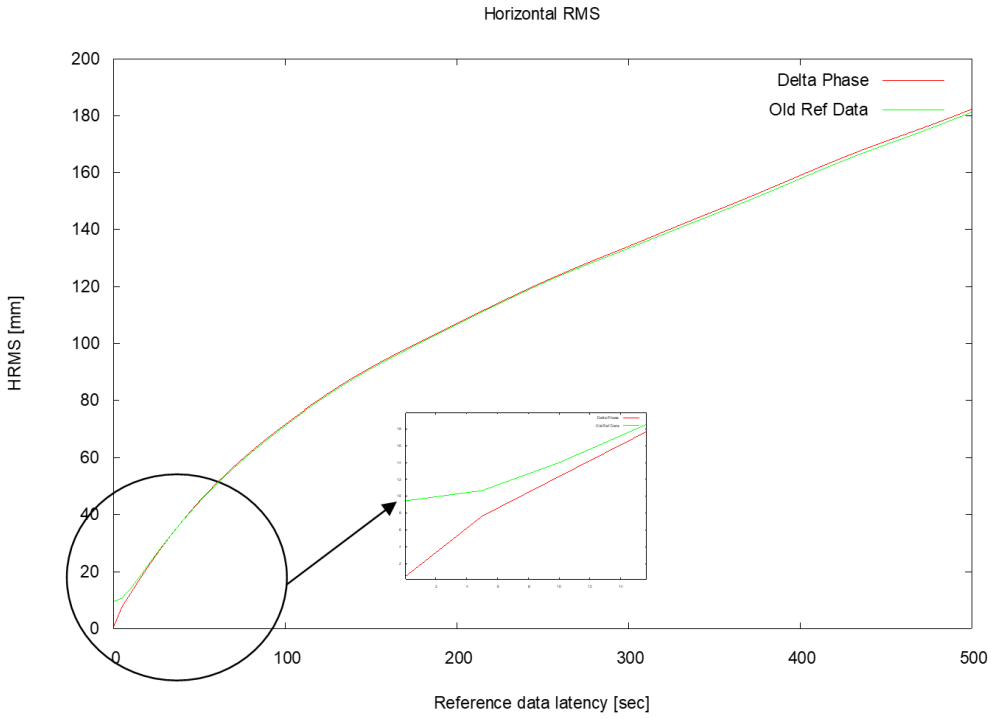


Figure 3: Comparison of "Delta Phase" processing with the approach using old reference data

Figure 2: Supplied excerpt headed *4.1 Delta-Phase Processing* [8], including its original text and graph. The early separation does not demonstrate a lasting advantage in absolute accuracy; the curves nearly coincide at longer latencies. Original publication metadata were not supplied.

A Minimal implementation checklist

1. Fix clock signs, phase units and signal identity.
2. Compute each observation's transmission time and complete satellite model.
3. Form new-minus-old clock changes with relativity exactly once.
4. Add $+cD_s$ to a raw temporal or asynchronous difference.
5. If adjusting the old reference side only, add $-cD_s$ there.
6. For a synthetic base epoch additionally predict geometry, atmosphere and the common receiver-clock gauge.
7. Preserve anchor covariance, shared-measurement correlations and arc validity.
8. Test deterministic signs before analysing real-data drift.
9. Validate horizontal/vertical errors against continuous-reference replay.

References

[1] J. Sanz Subirana, J. M. Juan Zornoza and M. Hernandez-Pajares. *GNSS Measurements Modelling*. ESA Navipedia, 2011. `https://gssc.esa.int/navipedia/index.php/GNSS_Measurements_Modelling`.

[2] J. Sanz Subirana, J. M. Juan Zornoza and M. Hernandez-Pajares. *Clock Modelling*. ESA Navipedia, 2011. `https://gssc.esa.int/navipedia/index.php/Clock_Modelling`.

[3] J. Sanz Subirana, J. M. Juan Zornoza and M. Hernandez-Pajares. *Relativistic Clock Correction*. ESA Navipedia, 2011. `https://gssc.esa.int/navipedia/index.php/Relativistic_Clock_Correction`.

[4] B. Shu, H. Liu, Y. Feng, L. Xu, C. Qian and Z. Yang. Analysis of Factors Affecting Asynchronous RTK Positioning with GNSS Signals. *Remote Sensing*, 11(10), 1256, 2019. `https://doi.org/10.3390/rs11101256`.

[5] J. Kim, Y. Kim, J. Song, D. Kim, M. Park and C. Kee. Performance Improvement of Time-Differenced Carrier Phase Measurement-Based Integrated GPS/INS Considering Noise Correlation. *Sensors*, 19(14), 3084, 2019. `https://doi.org/10.3390/s19143084`.

[6] NovAtel. *RTK ASSIST Services*. OEM7 documentation. `https://docs.novatel.com/OEM7/Content/Operation/RTK_ASSIST_Enabling.htm`. Manufacturer documentation; service performance is conditional.

[7] *SV-clock Synchronization during reference data outages*. User-supplied document, `2024-12-09.SV-clock-summary.docx`; notes dated 22 and 27 August 2024 and an addition dated 9 December 2024. Original publication metadata of its historical graph were unavailable. The document also contains GPS ICD extracts; use the applicable maintained constellation specification for production implementation.

[8] *Delta-Phase Processing*, section 4.1 and Figure 3. User-supplied image excerpt. Original author, publication title and date unavailable.